

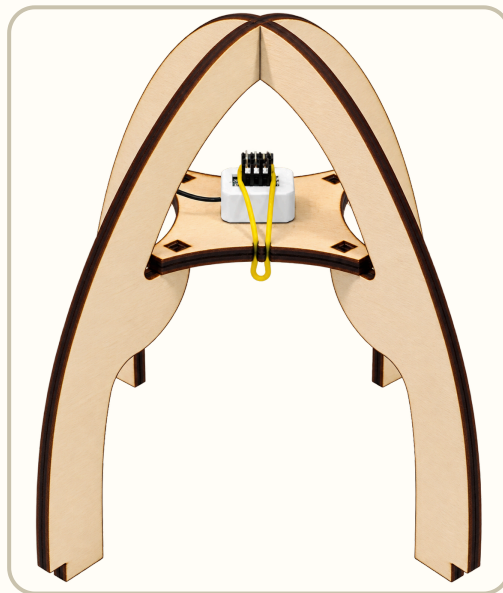


Bug Camera Day!

Today you are going to run a real insect camera, catch its pictures, and let a computer try to spot the bugs in them — all by yourselves.

A step-by-step guide for pupils and teachers · InsectAI Kit-for-Kids

GB English · ES [Español](#)



What you'll need

- A camera, SD card and Wi-Fi antenna
- The camera enclosure (white)
- The camera stage - made of three pieces of wood, two arches and one 4 pointed star piece
- An elastic band
- The little picture of ants that goes underneath the camera
- The battery pack
- A USB-C cable, to connect the battery pack to the camera
- A phone — **an Android phone works best** for today
- A laptop or computer to look at the pictures afterwards

Words to know

AI (Artificial Intelligence)

A computer program that has learned what insects tend to look like, and makes a best guess about where one might be in a photo. It can make mistakes — it's guessing, not certain!

SD card

The small memory card inside the camera that stores every photo it takes.

QR code

A special square pattern your phone's camera can read instead of you typing anything in.

Wi-Fi

How your phone talks to the camera without any wires, so you can peek at its photos while it's working.

Notes for the teacher / facilitator

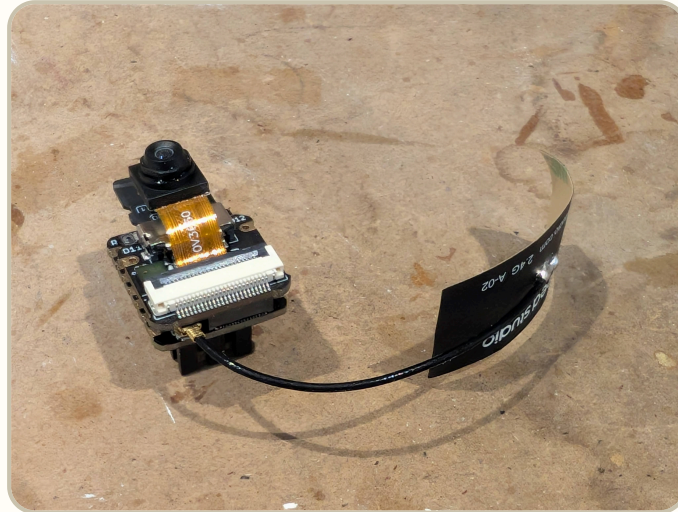
- Always wait for  **Finish my adventure** to confirm before removing the card — this is what safely saves the photos.
- Never remove the SD card while the camera is still powered on.
- Each camera has its own Wi-Fi name (InsectCam- plus a short label), so several cameras can run in the same room without clashing — check every camera is labelled with its name before the day starts.

Build your camera!

1

Connect the antenna and slot in the memory card

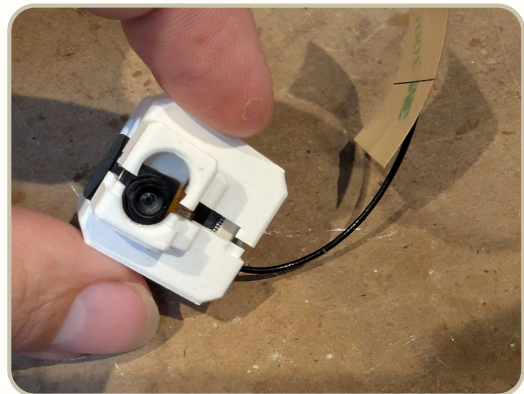
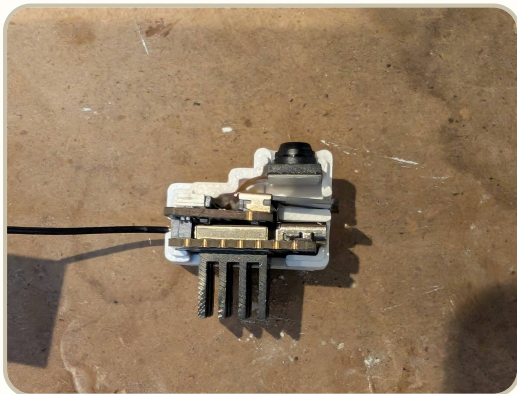
If the antenna and SD card are not already connected, plug the little Wi-Fi antenna onto the camera board, and slide the memory card into its slot. This is what the camera will use to talk to your phone and to save every picture it takes.



2

Close it into its case

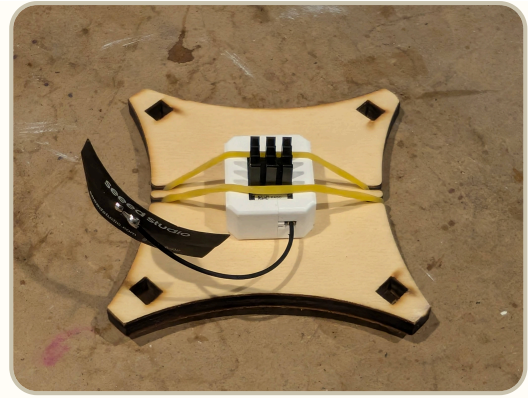
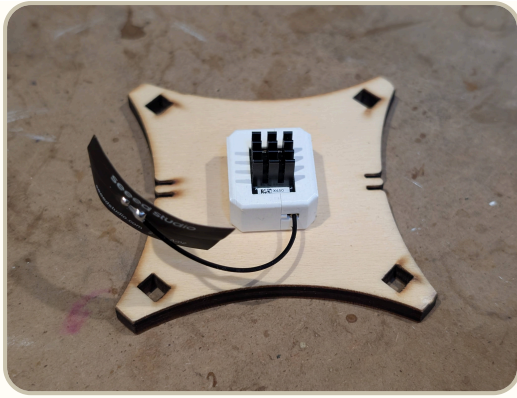
If the camera board is not already in the case, tuck it carefully into the white case so the lens looks out through the hole, then close the case.



3

Fix it to the star platform

Sit the camera into the wooden star-shaped platform so that the camera goes into the hole in the center. Stretch the elastic band across it, using the hooks on each side so that the elastic band goes around the heatsink, pressing the camera into the hole in the center.



4

Build the arch

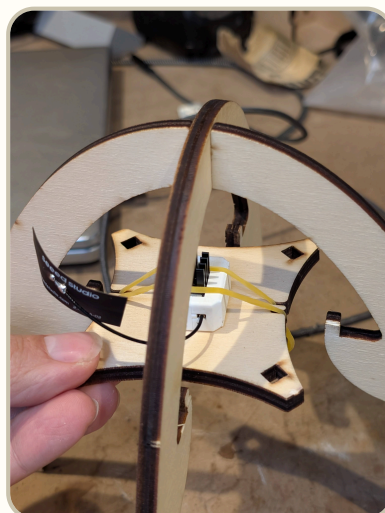
Slot the two curved wooden arch pieces together at the top, so they stand up together on their four legs.

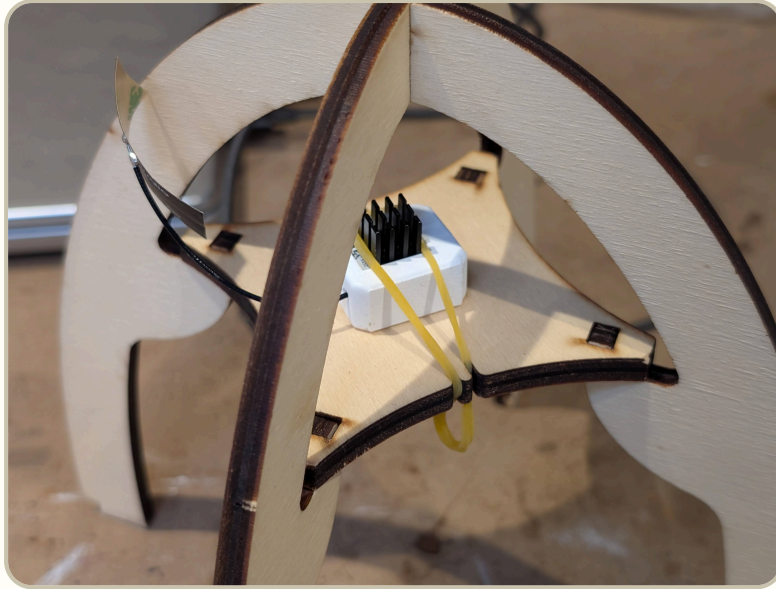


5

Slide the platform into the arch

Raise the star platform — with the camera strapped to it — up into arches. Once at the top of the arches rotate it and lower it onto the notches pressing it gently into place.





Great job — your camera is built! Now let's run it. 🎉

Let's go!

1

Put the picture of ants under the camera

This is what the camera will practice looking at. Line it up so it's nice and flat underneath the lens.



2

Plug in the battery

As soon as you connect the battery cable, the camera wakes up and starts taking a photo every second — all on its own!

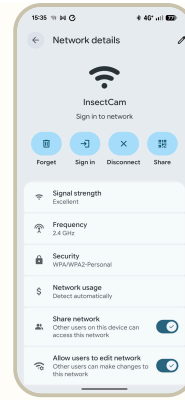
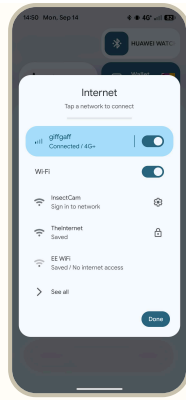


3

Join the camera's Wi-Fi with your phone

Open your phone's Wi-Fi settings and look for a network starting with InsectCam- — every camera has its own name, so check the label on your camera to find yours. Tap it, then type the password. A welcome page should open by itself. (If it doesn't, ask your teacher.)

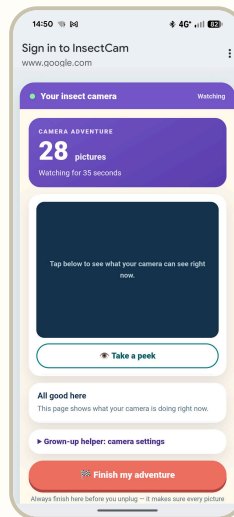
Network: InsectCam- ... (see your camera's label) Password: antcamera



4

Take a peek

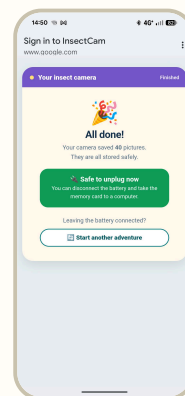
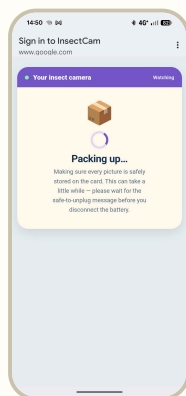
On the welcome page, tap  **Take a peek** to see the most recent photo the camera has just taken — live! Watch the pictures for about a minute while the camera keeps working.



5

Finish the adventure

When you're ready to stop, tap  **Finish my adventure**. Wait for it to say it's done — this makes sure all your photos are saved safely before you touch anything.

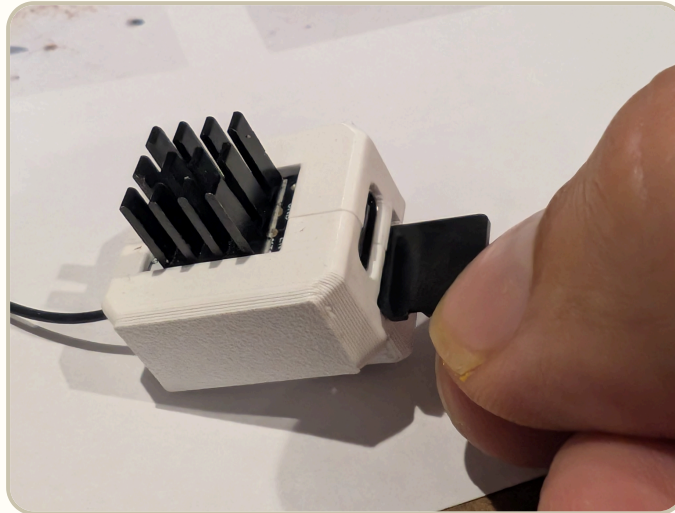


Now let's find the bugs!

6

Unplug and open up the card

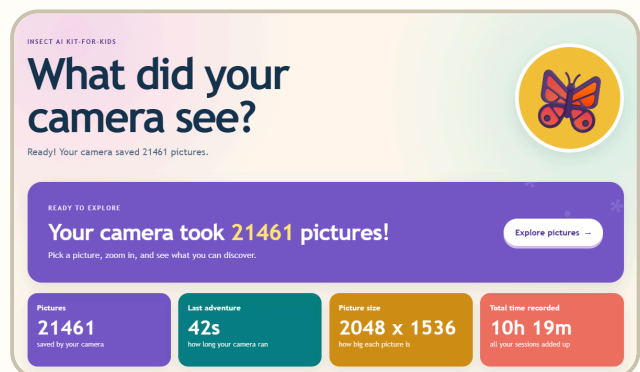
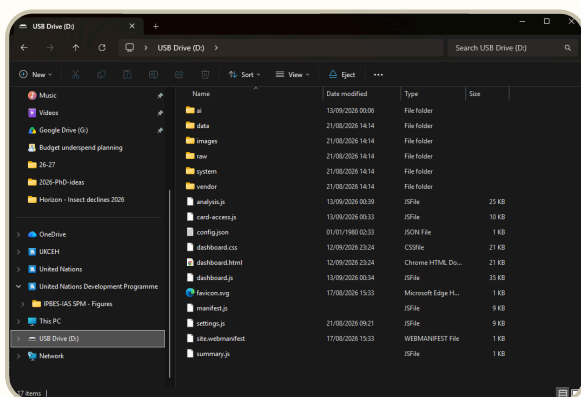
Disconnect the battery cable, carefully take out the memory card, and put it into a computer.



7

Open the picture gallery

On the computer, open the file called **dashboard.html** that's on the card — you don't need to install anything, just open it like any other file and it opens in your web browser. It shows every photo the camera just took — no internet needed, it all works right there on the card!



8

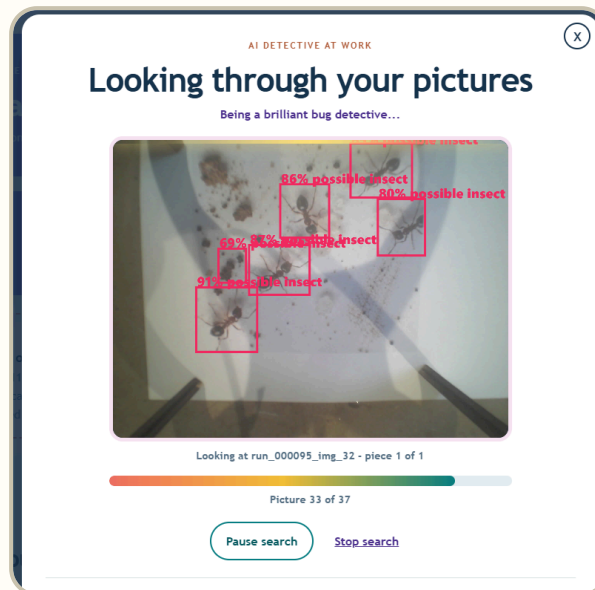
Load your pictures for the AI

Click **Find insects with AI**, then click **Load camera card** → . In the window that opens, choose your USB drive or SD card (look for **INSECT-AI**) and click **OK**. The browser will then ask if it's okay to look at all the files — click **OK** (or **Upload**) to allow it. Nothing is actually uploaded anywhere; this just lets the page on your computer read the pictures.

9

Let the AI look for insects

Pick **FlatBug — Quick look**, an AI model that has been trained to spot insects in photos. Watch as the computer looks at each photo, one at a time, and guesses where it thinks it can see an insect.



That's it — you've run a real insect-monitoring camera and used AI to look for bugs, start to finish! 🐛



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